

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

Project ID: 25-26J-103

Project Title: E-Learning platform for hearing impaired student

1. Introduction

1.1. Background:

Hearing impairment presents significant barriers to education, communication, and social participation. Approximately 9% of Sri Lankans experience some degree of hearing loss, while 15% are classified as profoundly deaf. These figures are projected to rise across South Asia from 131 million to 267 million by 2050. Despite these numbers, Sri Lanka remains underserved in terms of educational technology tailored to the hearing-impaired community. The COVID-19 pandemic further underscored the need for accessible online education, as many hearing-impaired students could not participate meaningfully in standard remote learning formats. Additionally, the exclusion of deaf individuals from cultural experiences such as music represents an important social and educational gap.

1.2. Research Problem:

Existing digital learning tools neither acknowledge nor incorporate Sri Lankan Sign Language (SSL) as a learning medium. Online platforms rely heavily on audio-based instruction, creating comprehension and communication barriers for hearing-impaired students. There is no dedicated lip-reading training platform for Sinhala phonemes, no real-time emotional engagement monitoring, no gesture evaluation module with real time feedback system for Sri Lankan e-learning, and no browser-native multi-sensory music accessibility platform for the deaf community in Sri Lanka.

1.3. Objectives:

It Number	Objective	Objective number
IT22546852	Develop an interactive Sri Lankan Sign Language (SSL) self-learning through 2D avatar and real video and gesture evaluation module using CNN and LSTM models with real-time feedback.	1
IT22143754	Create a lip-reading assistance module with a 3D avatar trained on a proprietary Sinhala lip movement dataset to support pronunciation training.	2
IT22561466	Build a teacher-student engagement module with second-by-second facial emotion detection and adaptive interactive activity interventions.	3

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IT22255488	Design and implement a multi-sensory deaf-first music experience platform (Resonance) integrating CNN emotion recognition, ESP32 haptic feedback, and AI-powered guidance.	4
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2. Data Exploration

2.1. Data Collection

Data for the Hear Me platform was collected from multiple sources to address the four distinct research objectives. The SSL gesture recognition module utilizes the publicly available SSL400 Kaggle dataset of dynamic Sri Lankan Sign Language video sequences. The lip-reading module required a proprietary dataset collected by recording multiple participants articulating curated Sinhala words and sentences under controlled lighting conditions. The emotion detection module leverages the DeepFace library trained on large-scale public facial expression datasets. The Resonance music platform utilises audio files processed through librosa for feature extraction, combined with Google Cloud Speech-to-Text for real-time lyrics transcription. Additionally, a custom static hand-gesture image dataset was captured via webcam using OpenCV, with approximately 300–400 grayscale images per class.

2.2. Dataset Description:

Data source	Description	Resource	Size	Key attributes
SSL400 Dataset	Dynamic Sri Lankan Sign Language video sequences covering 400 word classes recorded by multiple signers.	Kaggle (Public)	400 classes	Video sequences, multi-signer recordings, SSL vocabulary; MediaPipe landmark extraction.
Custom SSL Landmark Dataset	Frame-by-frame hand, pose, and face landmarks extracted from SSL400 videos using MediaPipe Holistic.	Custom (derived from SSL400)	40 seq x 30 frames	NumPy arrays of hand/pose/face landmarks, temporal sequences, categorical labels.
Custom Static Hand Image Dataset	Grayscale webcam-captured hand	Custom (OpenCV)	300–400 images/class	64x64 px grayscale images, class

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	gesture images for CNN-based static sign classification.	webcam capture)		labels, gesture categories.
custom Sinhala lip-reading dataset	Proprietary dataset of participants articulating Sinhala words and sentences captured under controlled conditions.	Custom (proprietary)	Multiple participants, curated word set	Lip region image sequences, phoneme labels, facial landmark annotations, Sinhala text mappings.
Facial Expression Dataset (DeepFace)	Pre-trained deep learning model for facial emotion classification across standard expression categories.	Public (DeepFace library)	Large-scale (pre-trained)	Facial images, emotion labels (happy, neutral, confused, bored, disengaged), 97% detection accuracy.
Audio Feature Dataset (Resonance)	Audio files processed into mel-spectrograms for CNN-based music emotion recognition using librosa.	Processed audio (custom pipeline)	5-second chunks, 44.1 kHz	28-band mel-spectrograms, valence/arousal labels, Russell Circumplex Model mappings.

2.3. Suitability Analysis

2.3.1. Relevance to Individual Research Objectives:

(Explain how well each dataset aligns with your research problem and objectives)

	1	2	3	4
SSL400 Dataset	Highly relevant (primary vocabulary and video source for gesture recognition training.)	Not applicable.	Not applicable.	Not applicable.
Custom SSL Landmark Dataset	Directly used for LSTM temporal	Not applicable.	Not applicable.	Not applicable.

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	sequence training.			
Custom Static Hand Image Dataset	Supports CNN-based static sign classification within LSTM pipeline.	Not applicable.	Not applicable.	Not applicable.
Sinhala Lip Movement Dataset	Not applicable.	Primary training data for CNN-LSTM lip-reading model; essential for Sinhala phoneme coverage.	Not applicable.	Not applicable.
Facial Expression Dataset (DeepFace)	Not applicable.	Not applicable.	Highly relevant (provides 97% face detection and ~80% emotion classification accuracy for real-time monitoring.)	Not applicable.
Audio Feature Dataset (Resonance)	Not applicable.	Not applicable.	Not applicable.	Highly relevant (mel-spectrogram features directly drive CNN emotion recognition for music accessibility.)

3. Methodology

3.1. Data Preprocessing:

(Mention data transformation techniques done in each dataset for each objective.)

Ex:

Data Cleaning, Data Normalization, Data Standardization, Data Encoding (e.g., One-Hot Encoding, Label Encoding), Handling Missing Data (e.g., Imputation or Removal), Data Aggregation, Feature Engineering, Outlier Detection and Handling, Data Scaling, Data Discretization, Dimensionality Reduction (e.g., PCA), Date/Time Transformation, Data Integration (Merging or Joining), Data Mapping, Data Type Conversion.

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Objective 1

Data Source	Transformation Technique
SSL400 / Custom Landmark Dataset	MediaPipe Holistic landmark extraction; feature concatenation (hand + pose + face) per frame; NumPy array serialization; frame normalization to 30 frames/sequence; categorical label encoding; train/validation split.
Custom Static Hand Image Dataset	Grayscale conversion (cv2.cvtColor); resizing to 64x64 px (cv2.resize); pixel normalization [0,1]; data augmentation; label encoding; train/test split.

Objective 2

Data Source	Transformation Technique
Sinhala Lip Movement Dataset	Lip region extraction via MediaPipe Face Mesh; bounding box normalization; frame sequence standardization; phoneme-level label mapping; dropout regularization; early stopping applied during training.

Objective 3

Data Source	Transformation Technique
Facial Expression Dataset (DeepFace)	Real-time webcam frame capture; facial detection preprocessing; emotion classification every 1 second; per-student state tracking; session-level aggregation across all students.

Objective 4

Data Source	Transformation Technique
Audio Feature Dataset	Audio resampling to 44,100 Hz mono (librosa); 5-second chunk segmentation; 128-band mel-spectrogram generation (n_fft=2048, hop_length=512); decibel-scale conversion; tensor reshaping to (128 x T x 1); valence/arousal denormalization using training-set min/max statistics.
Transcription	Audio conversion to 16 kHz mono 16-bit PCM WAV via FFmpeg; Sinhala Unicode range validation (U+0D80–U+0DFF, 70% threshold);

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	whitespace normalization; removal of non-Sinhala characters.
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3.2. Scalability

The SSL400 dataset with 400 word classes provides a scalable vocabulary foundation; additional SSL word categories can be incorporated by appending new video sequences and retraining the LSTM pipeline. The custom hand image dataset (300–400 images per class) is sufficient for initial CNN training, with scalability supported by OpenCV-based automated capture. . The proprietary Sinhala lip dataset is the most constrained in size; scalability depends on recruiting additional participants to increase phoneme coverage and model generalisation. The DeepFace-based emotion model scales well across concurrent students, with per-student webcam streams processed independently. The Resonance audio pipeline scales horizontally using FastAPI microservices, with health polling ensuring service stability.

3.3. Feature extraction

- Object 01

SSL: MediaPipe Holistic extracts 21 hand landmarks + 33 pose landmarks + 468 face landmarks per frame; concatenated into a fixed-length feature vector per timestep for LSTM input. CNN extracts spatial hierarchical features from 64x64 grayscale hand images through convolutional and pooling layers.

- Object 2

Lip-Reading: MediaPipe Face Mesh isolates the lip region per frame; spatial features extracted via CNN convolutional layers; temporal motion trajectory across frames captured by LSTM layers.

- Object 3

Emotion: DeepFace extracts facial bounding box and emotion probabilities per second; aggregated emotional state vectors computed across the session for teacher dashboard reporting.

- Object 4

Resonance: librosa extracts 128-band mel-spectrograms per 5-second audio chunk as the primary CNN input; RMS energy computed per frame for haptic onset detection; FFT frequency bins mapped to individual vibration motors.

4. Modelling and Results

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The results section should present the key findings derived from analyzing the data. It should answer the research questions or hypotheses posed at the beginning of the study, based on the methods and techniques applied during the analysis.

4.1. Key Insights:

Objective 1

The LSTM-based dynamic sign classifier trained on SSL400 landmark sequences achieved strong performance. Classification accuracy was inversely correlated with the number of input classes and positively correlated with inter-gesture distinctiveness. The CNN frame-level sub-classifier improved accuracy for static and near-static signs.

Category	Sign Level	Model	Accuracy	F1 Score
1 (Static)	Colors (9 classes)	CNN	87.8%	0.87
1 (Static)	Numbers 1–10	CNN	85%	0.85
2 (Dynamic)	Family Members (3)	LSTM	85%	0.85

User testing confirmed that dual-format demonstration (real video alongside 3D avatar) significantly improved confidence in sign execution. Per-gesture feedback for Category 2 signs was particularly valued by participants.

Objective 2

The CNN-LSTM classifier evaluated on the held-out Sinhala lip dataset demonstrated the effectiveness of combining convolutional spatial feature extraction with LSTM temporal modeling for phoneme-level lip classification.

Model	Accuracy	Loss Value
CNN (Lip-Reading Classifier)	67%	0.124

Multi-angle avatar playback significantly improved understanding of unfamiliar phoneme articulations compared to single-angle presentation. Slowing avatar playback was particularly effective for mastering complex phoneme transitions

Objective 3

The emotion detection subsystem classified student emotional states at 1-second intervals achieving 97% face detection accuracy and approximately 80% emotion classification accuracy under standard webcam conditions. The 20-second adaptive

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activity intervention measurably increased re-engagement in pilot sessions. Teacher feedback indicated the system did not disrupt lesson delivery, and the emotion monitoring dashboard provided actionable insights into student engagement patterns

Objective 4

The CNN emotion recognition model achieved consistent inference latency under 200ms per 2-second audio chunk, enabling near real-time emotion tracking. The onset-based haptic algorithm was unanimously preferred over simple amplitude mapping by all hearing-impaired users in the evaluation, producing distinct, meaningful beat sensations. The Sinhala Unicode validation filter successfully rejected all romanized false-positive transcriptions during testing.

Metric	Value
CNN Emotion Inference Latency	<200ms per 2-second audio chunk
Haptic Output Refresh Rate	~30 fps (33ms interval)
Supported Transcription Languages	11 (including Sinhala -LK)
Sinhala Unicode Validation Threshold	70% character dominance (U+0D80–U+0DFF)

4.2. Challenges Faced During Data Analysis:

- SSL400 Dataset

Variability: The multi-signer nature of SSL400 introduced inter-signer variation in hand shape, speed, and signing angle, requiring robust landmark normalization and sufficient sequence length to achieve consistent classification.

- Sinhala Lip Dataset

Limited Size: The proprietary Sinhala lip dataset is constrained by participant availability and controlled recording conditions, limiting model generalisation to unseen speakers. Dropout regularization and early stopping were applied to mitigate overfitting.

- Real-Time Emotion Detection

Webcam Conditions: Emotion classification accuracy was sensitive to ambient lighting and webcam quality, with performance degrading under poor lighting. The 97% face detection rate assumed adequate webcam conditions.

- Resonance

Transcription False Positives: Google Cloud Speech-to-Text produced romanized false-positive transcriptions for Sinhala audio, necessitating the implementation of a character-level Unicode validation filter requiring at least 70% Sinhala character dominance.

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Haptic Hardware Latency: Synchronising browser-side Web Serial API communication (~30 fps) with the ESP32 PWM haptic output required careful tuning of frame-to-frame energy delta thresholds and motor inertia compensation (minimum PWM threshold of 80).

Computational Limits: Running simultaneous webcam-based gesture recognition, facial emotion detection, and audio processing on standard consumer hardware required careful optimization of inference pipelines to maintain real-time responsiveness across all components.

5. References

- 1) DailyFT, "Sinhala Sign Language the main communication mode for the Deaf in Sri Lanka," 2019. [Online]. Available: <https://www.ft.lk/Opinion-and-Issues/Sinhala-Sign-Language-the-main-communication-mode-for-the-Deaf-in-Sri-Lanka/14-671078>
- 2) J. Elfein, "Number of people with hearing loss worldwide," Statista, June 2020. [Online]. Available: <https://www.statista.com/statistics/888654>
- 3) P. Haththotuwa and R. Rupasinghe, "Adapting to Online Learning in Higher Education System during the Covid-19 Pandemic: A Case Study of Universities in Sri Lanka," Sri Lanka Journal of Social Sciences and Humanities, vol. 1, no. 2, p. 147, 2021.
- 4) W. Aljedaani et al., "I Cannot See You—The Perspectives of Deaf Students to Online Learning during COVID-19 Pandemic: Saudi Arabia Case Study," Education Sciences, vol. 11, no. 11, p. 712, 2021.
- 5) D. Manoj Kumar et al., "EasyTalk: A Translator for Sri Lankan Sign Language using Machine Learning and Artificial Intelligence," in Proc. 2020 2nd Int. Conf. Advancements in Computing (ICAC), 2020, pp. 506-511.
- 6) S. M. Dissanayake et al., "Utalk: Sri Lankan Sign Language Converter Mobile App using Image Processing and Machine Learning," in Proc. 2020 2nd Int. Conf. Advancements in Computing (ICAC), 2020, pp. 31-36.
- 7) H. Amnur, Y. Syanurdi, R. Idmayanti, and A. Erianda, "Developing Online Learning Applications for People with Hearing Disabilities," JOIV: International Journal on Informatics Visualization, vol. 5, no. 1, 2021.
- 8) N. Krishnamoorthy et al., "E-Learning Platform for Hearing Impaired Students," in Proc. 2021 3rd Int. Conf. Advancements in Computing (ICAC), 2021, pp. 122-127.
- 9) D. Wanasinghe et al., "Hashta: Online Learning Platform for Hearing Impaired," in Proc. 2022 Int. Conf. on ICT for Emerging Regions (ICTer), 2022.
- 10) C. Dotty, "Sentiment Analysis and Emotion Recognition: What's the Difference?" Deepgram Blog, 2022. [Online]. Available: <https://deepgram.com/blog/sentiment-analysis-emotion-regulation-difference>
- 11) Resonance Project, "Resonance: A Deaf-First Music Experience Platform—Comprehensive Technical Documentation," Technical Reference v1.0, March 2026.
- 12) Y. Abhishek, "SSL400 – Dynamic Sri Lankan Sign Language Dataset," Kaggle, 2023. [Online]. Available: <https://www.kaggle.com/datasets/yohanabhishek/ssl400-dynamic-sri-lankan-sign-language-dataset>